



Enhancing Breast Cancer Detection in Mammography

ResNet50, ResNet101, EfficientNet,
Vit-Base, and Vit-Medium

Michael Sanfilippo(mds9427) - Matthew Zhou (hz3463)

<https://github.com/Mattz-CE/rsna>

Why Breast Cancer Detection in Mammography?

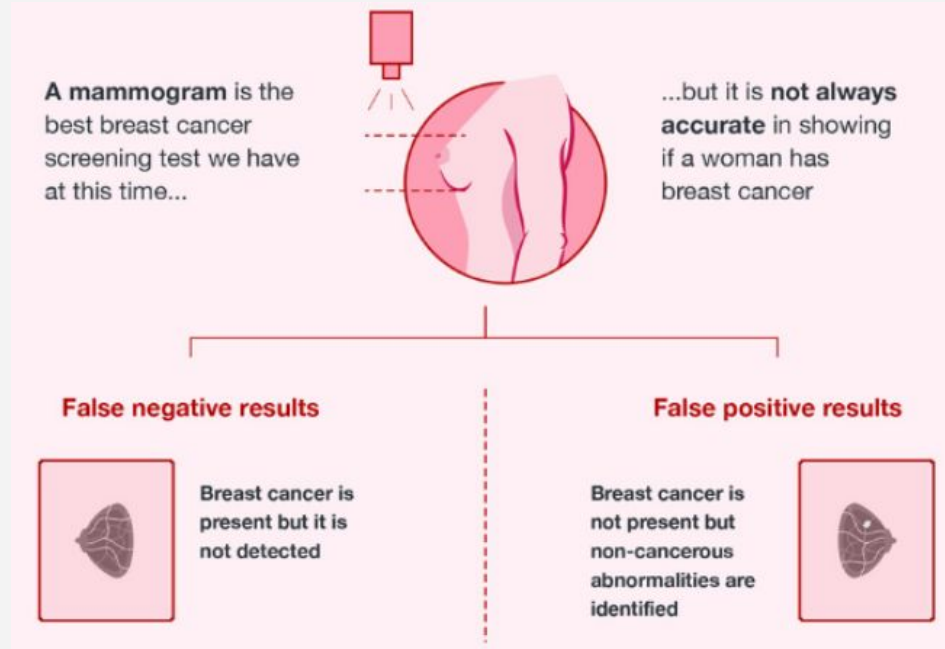


Figure 1: Mammogram False Positives and Negatives



DATASET

RSNA Screening Mammography
Breast Cancer Detection



Figure 2: Mediolateral Oblique Orientation Mammogram

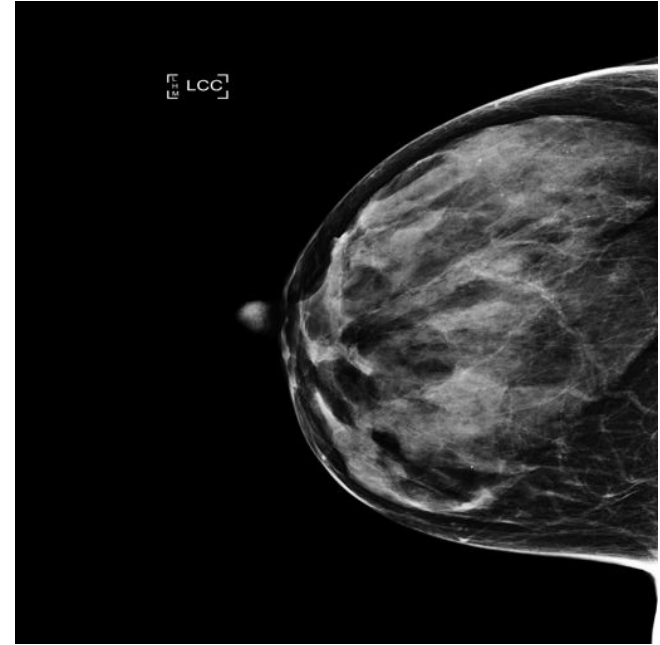


Figure 3: Craniocaudal Orientation Mammogram

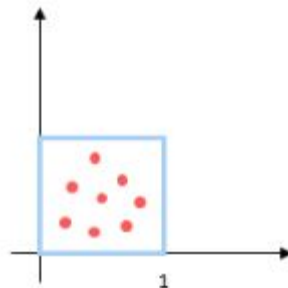
Preprocessing



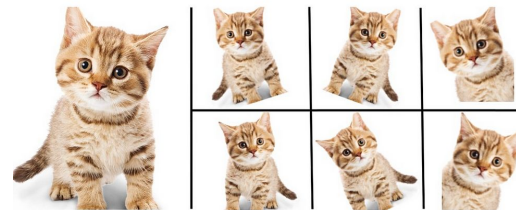
File Format
Conversion



Resizing

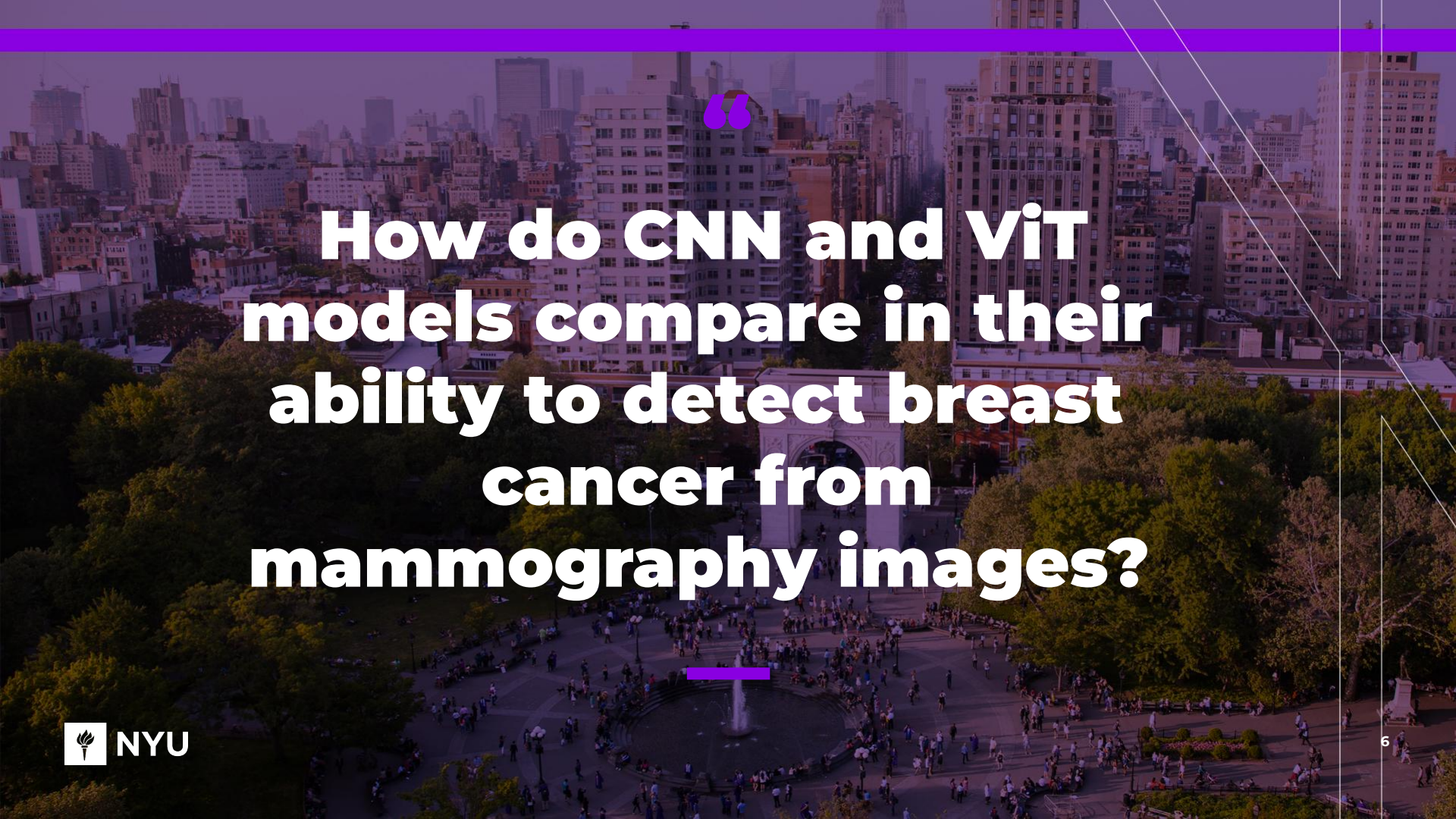


Normalizing



Enlarge your Dataset

Data Augmentation

An aerial photograph of Central Park in New York City, showing the Bethesda Fountain and the surrounding dense urban landscape of Manhattan. The image is overlaid with a semi-transparent purple gradient and white text.

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How do CNN and ViT models compare in their ability to detect breast cancer from mammography images?

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Hypothesis

Vision Transformers (ViT-Base and ViT-Medium) will outperform traditional CNNs (ResNet-50, ResNet-101, and EfficientNet) in higher AUC scores.



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Methodologies

Training Infrastructure, Models,
Hyper-Parameter Searching,
Metrics

Infrastructure

- Model checkpoints
- TensorBoard visualization
- Script versioning
- Early Stopping (with patience of 3 epoch)
- Distributed training across available GPUs
- Cloud GPU provided by Runpod.io

ResNet-50 & ResNet-101

Residual Networks

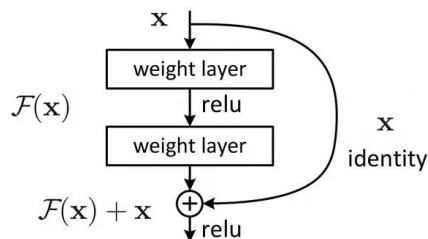


Figure 4: Residual Network Architecture Diagram

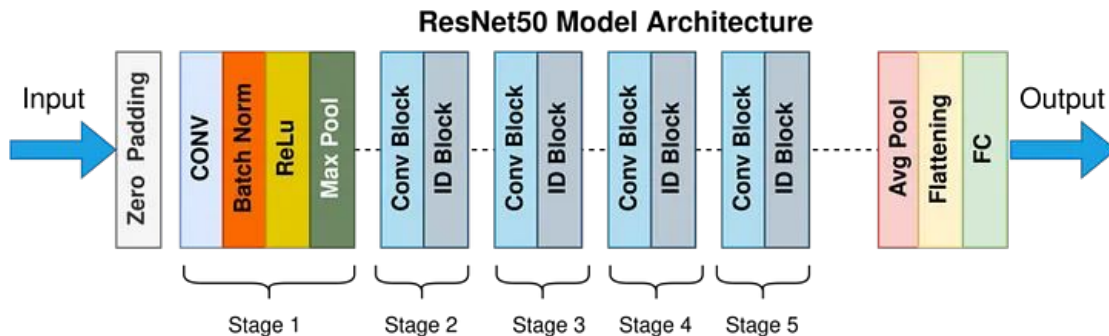


Figure 5: ResNet-50 Model Architecture

EfficientNet

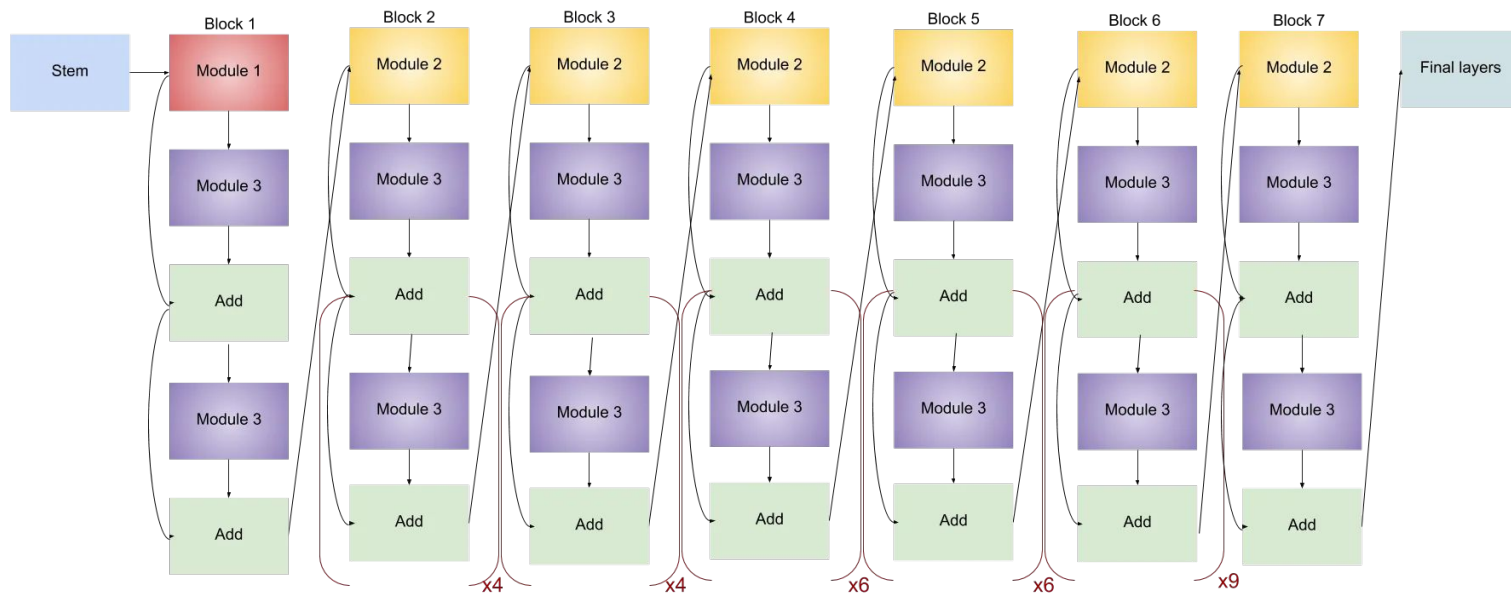


Figure 6: EfficientNet-B6 Architecture Diagram

ViT-Base & ViT-Medium

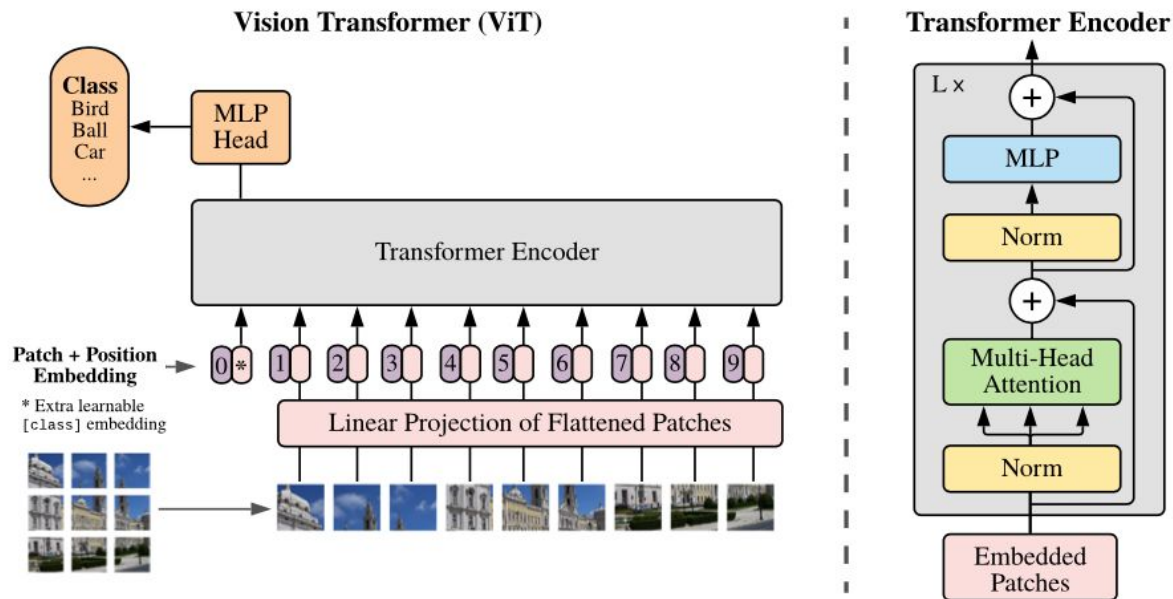


Figure 7: Visual Transformer (ViT) Architecture Diagram

Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2020). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv preprint*. <https://arxiv.org/abs/2010.11929>

Foundational vs Fine-Tuning

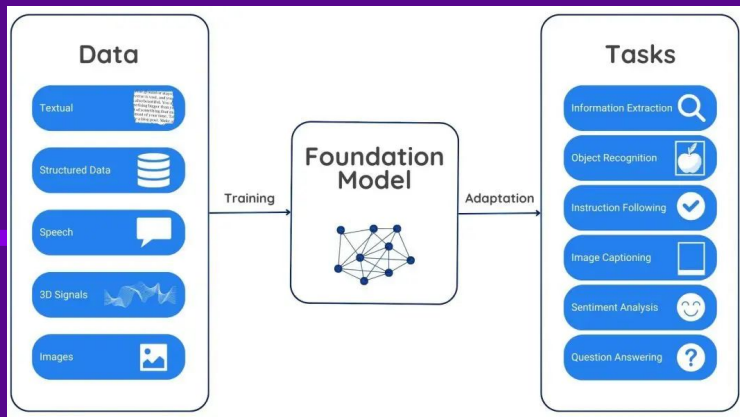


Figure 8: Foundational Training

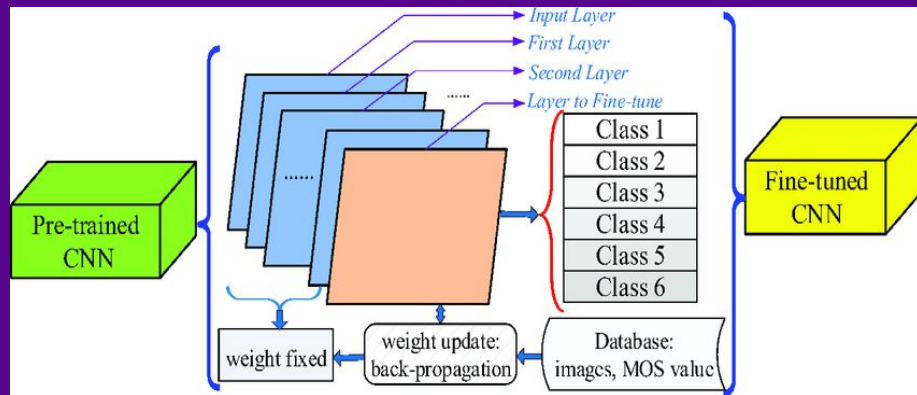


Figure 9: Fine-Tuning

HYPERPARAM SEARCHING

```
1  {
2      "model_name": ["resnet", "resnet101", "effnet", "vit_base", "vit_mediumd"],
3      "batch_size_per_gpu": [8,16,32,48,64],
4      "patch_size": [14,16,28],
5      "num_workers": [2,4,6,10,12,16],
6      "unfreeze_layers": [1,2,3,4,5],
7      "img_size": 512,
8      "epochs": 35,
9      "learning_rate": [0.002, 0.001, 0.005, 0.0001],
10     "validation_split": 0.2,
11     "early_stopping_margin": 0.03,
12     "early_stopping_patience": 3
13 }
```

Figure 10: Hyperparameter values for various models

Metrics Overview



AUC, ACC, LOSS

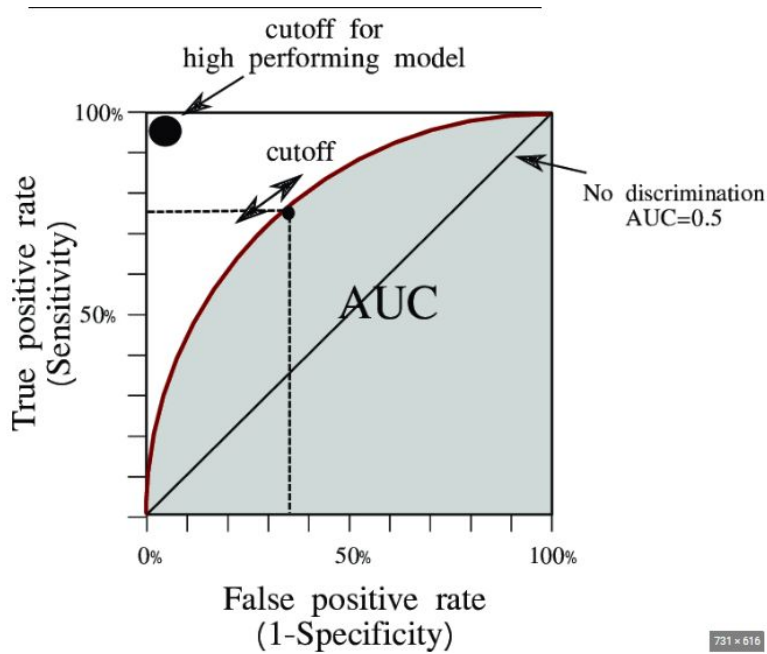
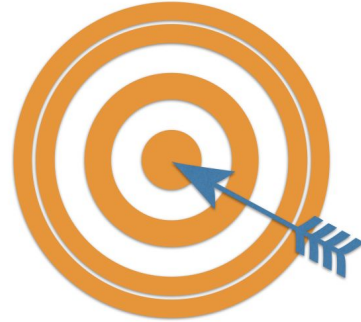


Figure 11: Area Under Curve

CLASSIFICATION ACCURACY, EXPLAINED



$$\frac{TP + TN}{TP + TN + FP + FN}$$

Figure 12: Accuracy Metric



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Results



ResNet-50

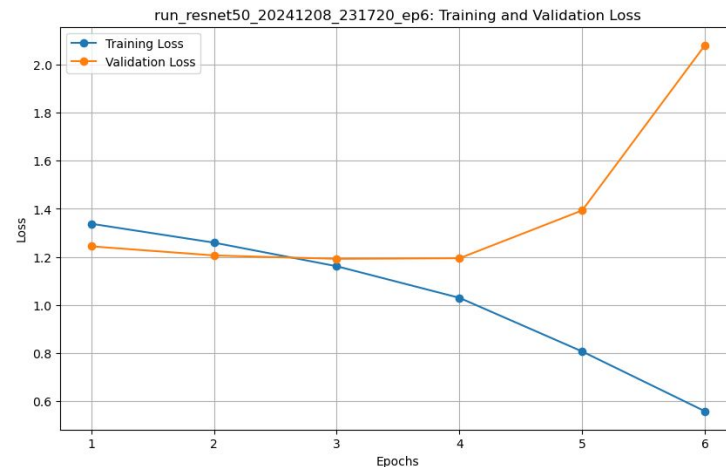
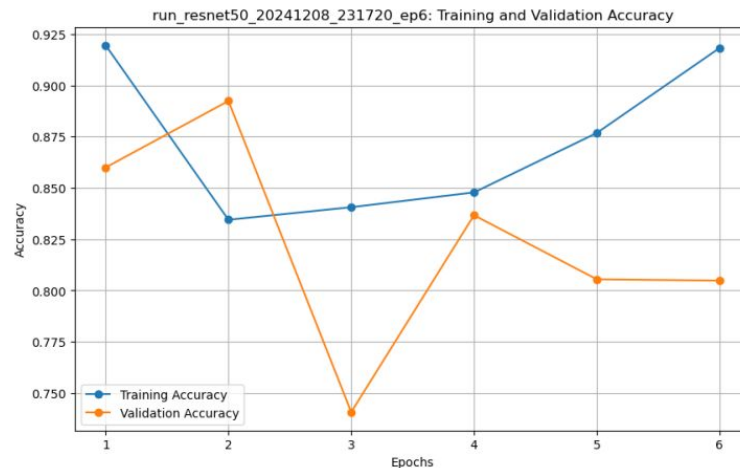
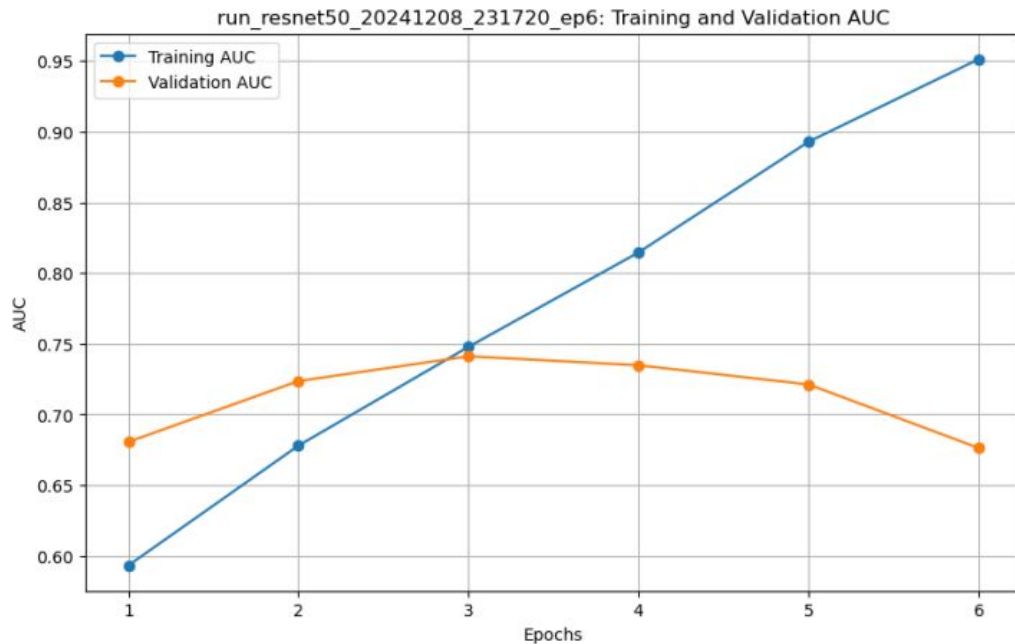
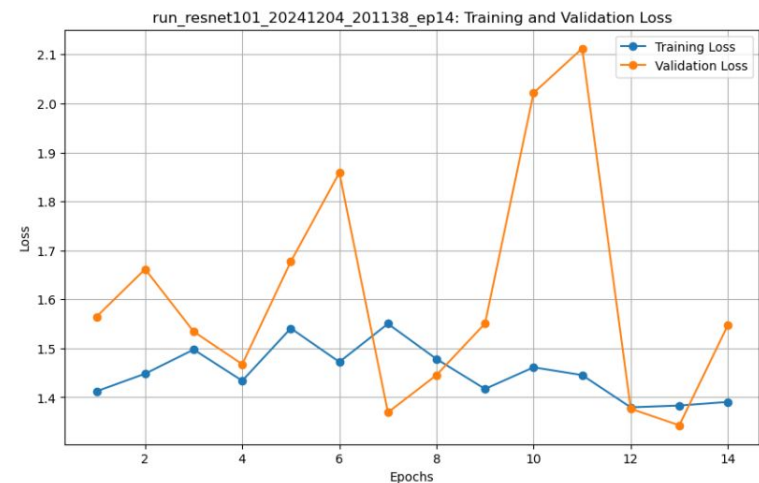
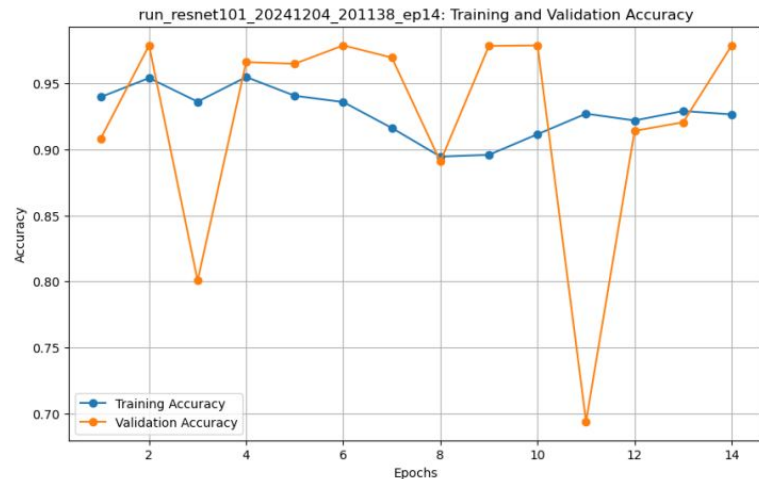
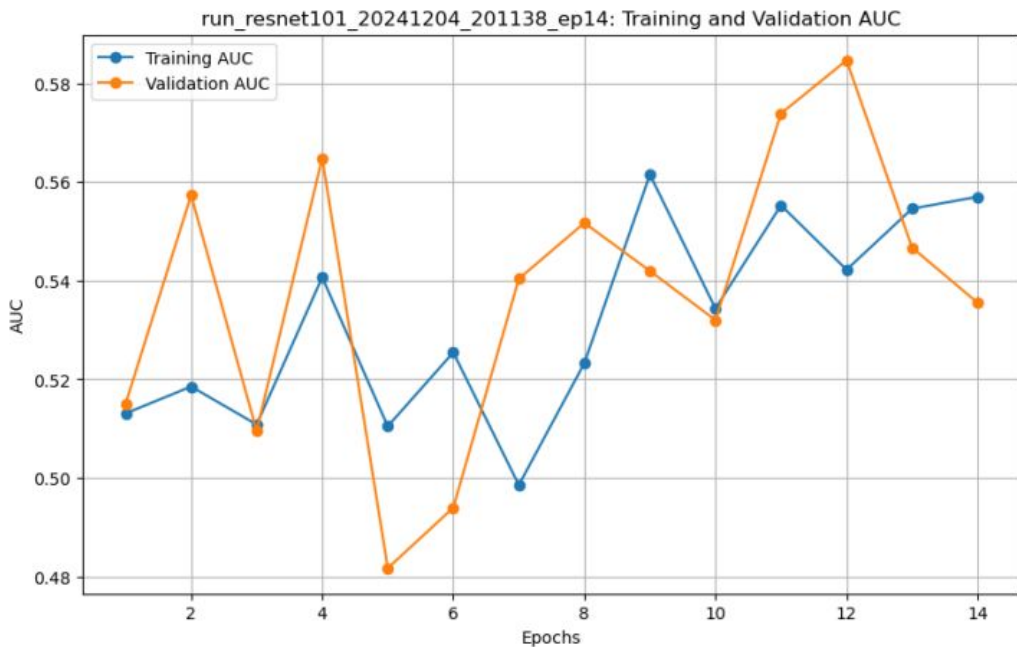
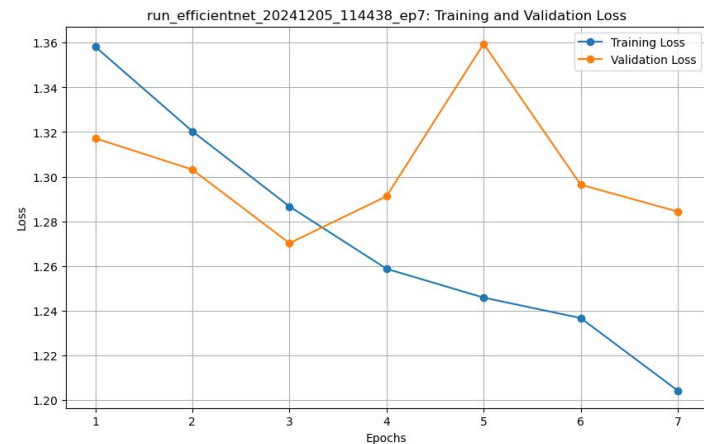
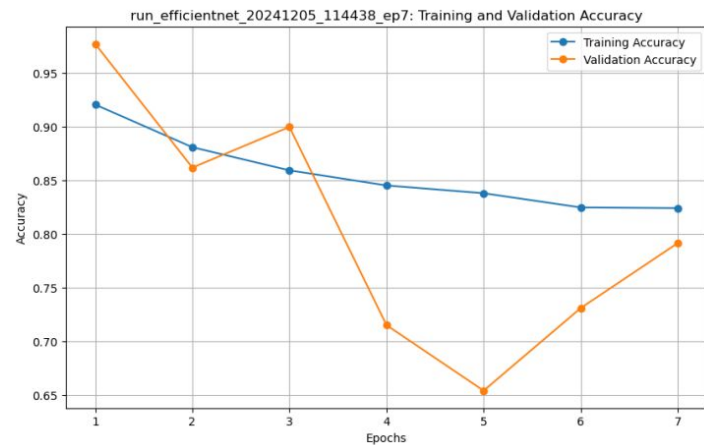
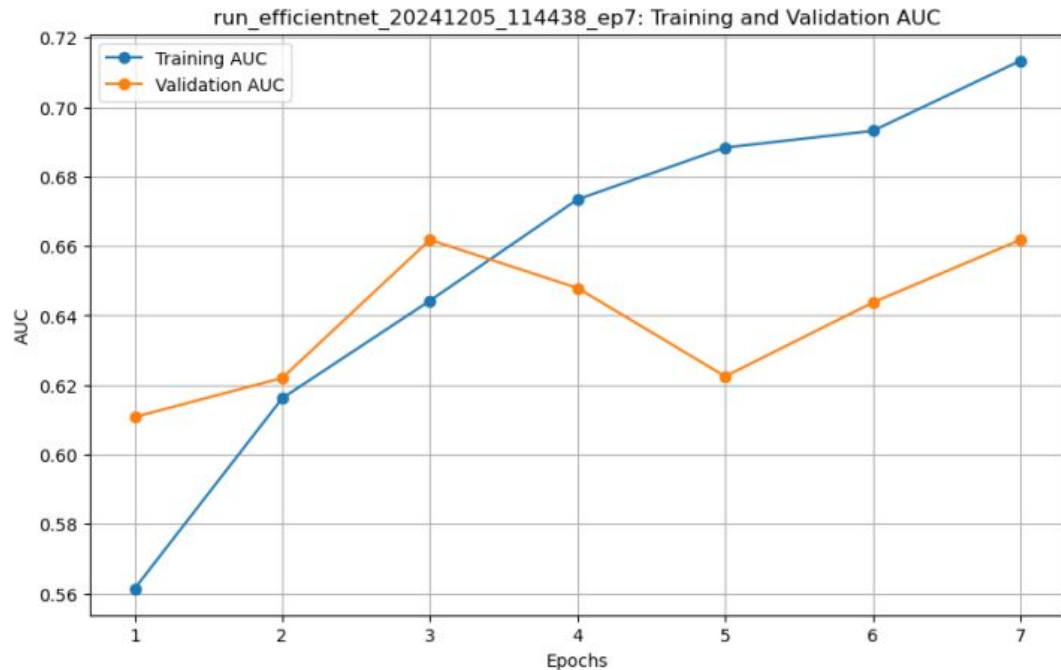


Figure 19, 20, 21: ResNet-50 AUC, Accuracy, and Loss Graphs

ResNet-101



EfficientNet



ViT-Base

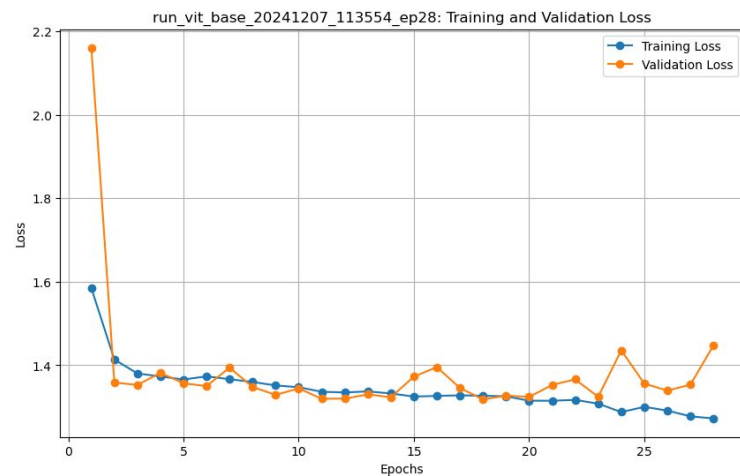
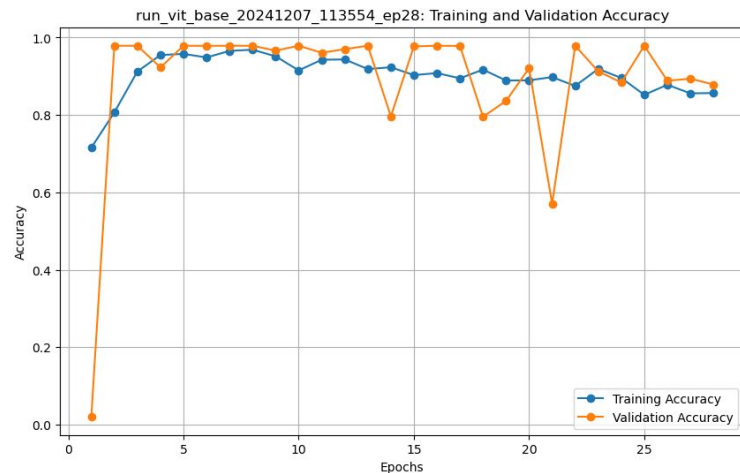
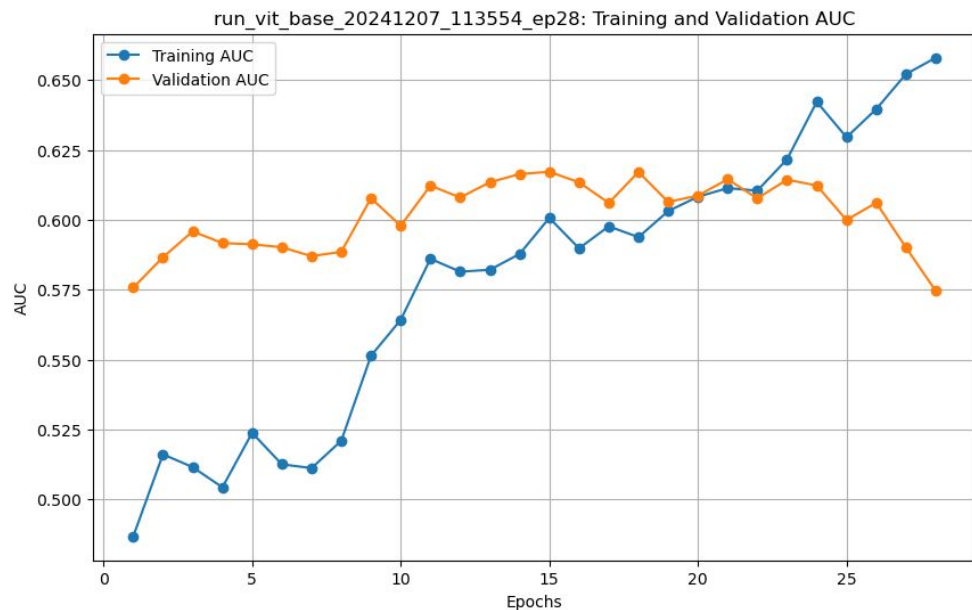


Figure 13, 14, 15: ViT Base AUC, Accuracy, and Loss Graphs

ViT-Medium

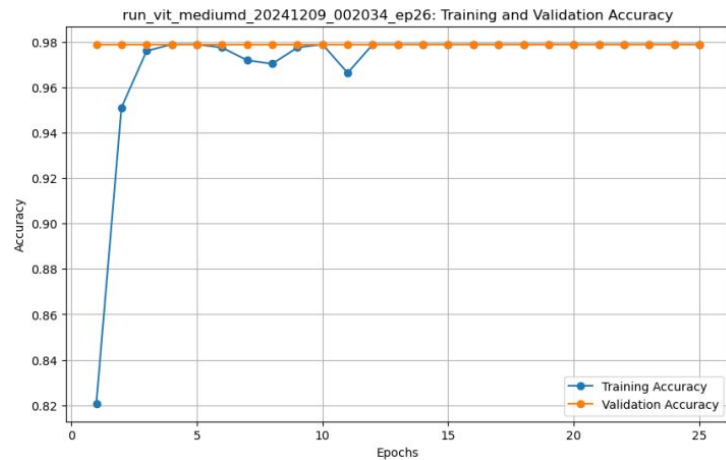
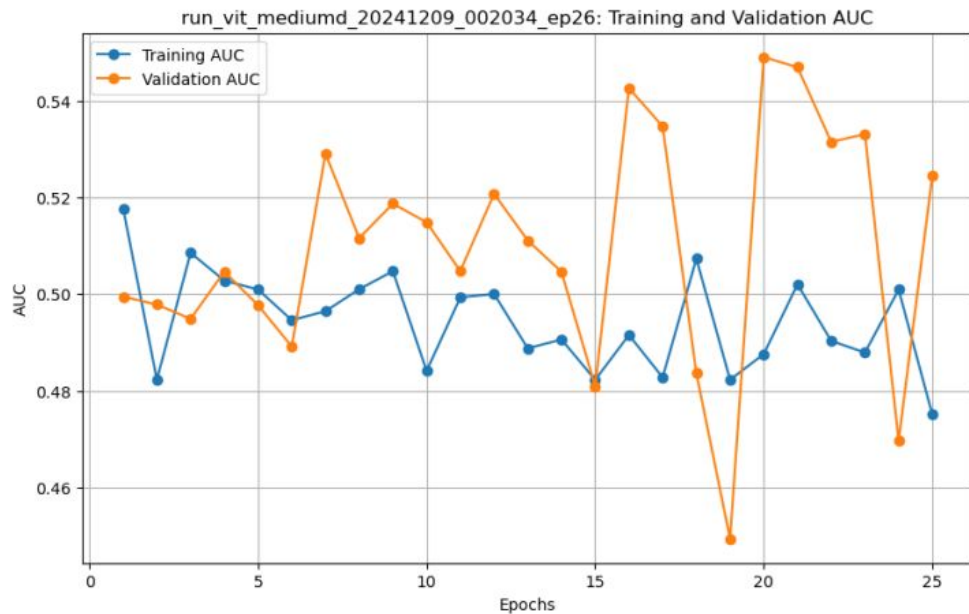


Figure 16, 17, 18: ViT Medium AUC, Accuracy, and Loss Graphs

Model Comparison

	Model	Best val_accuracy	Best val_auc	Best val_loss	Epoch	Total parameters
1	resnet50	0.740632	0.741148	1.191993	6	23503809
2	efficientnet	0.791354	0.661808	1.284357	7	20178337
3	vit_base	0.977244	0.617185	1.372631	28	86042113
4	resnet101	0.914092	0.584653	1.376719	14	42495937
5	vit_mediumd	0.978797	0.549083	1.358324	25	63727617

Figure 28: Comparison of Results for Different ML Models

74.1%

Best AUC Score Model

RESNET50

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**Our hypothesis was
disproved:
Larger models don't
necessarily mean
better performance...**

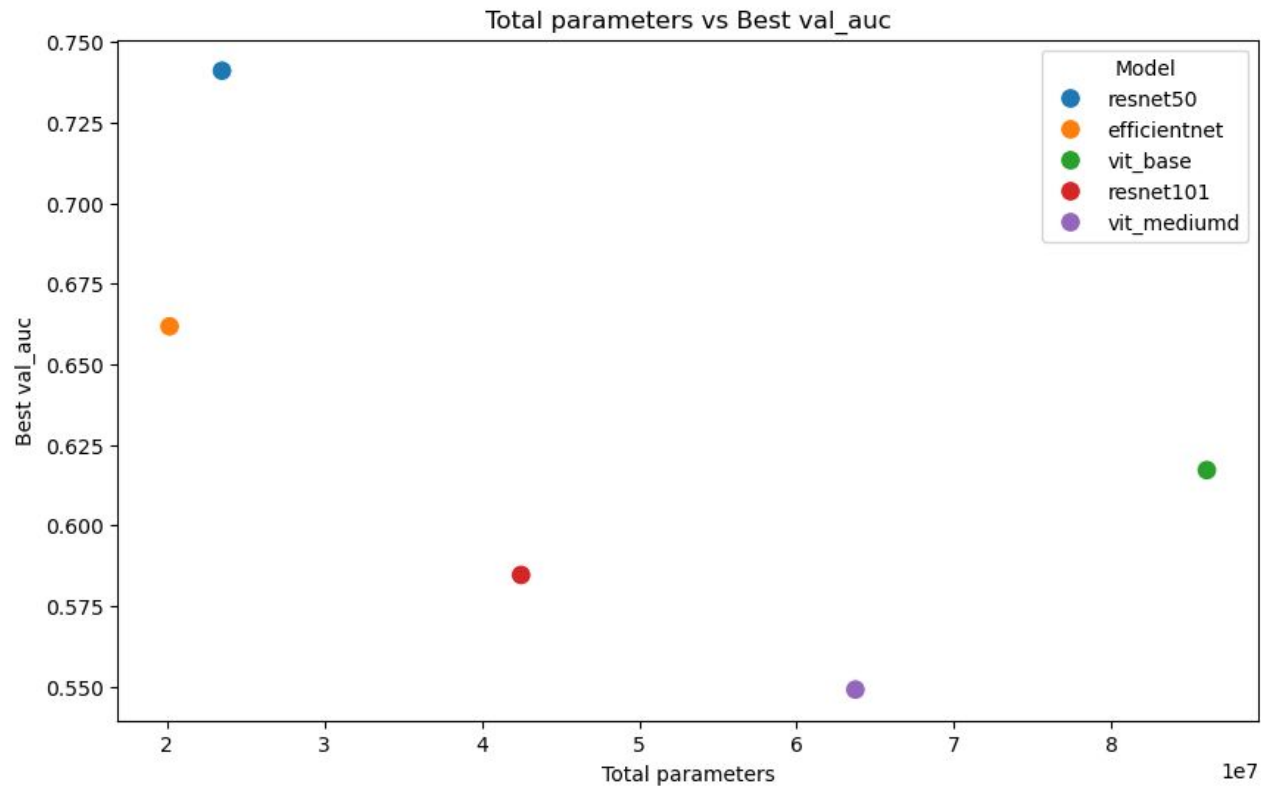


Figure 29: AUC relation to Model Size

Potential Reasons

1. Imbalance due to an overwhelming number of negative cases
2. Overfitting caused by larger model architectures



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QUESTIONS?

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